Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

AI Engineer with 8+ years of production Python engineering, specializing in **LLM evaluation infrastructure, agentic systems, and production observability**. Built AgentProbe — an open-source agentic AI testing toolkit with a unified 5-metric LLM-as-judge eval engine (factual accuracy, hallucination, retrieval quality, completeness, compliance), trace-level agent observability, automated CI/CD quality gates, and multi-provider LLM routing. Built LeaseI — an agentic RAG compliance pipeline with a pass@k evaluation harness (pass@1 81.7%, pass@3 88.3%) and LLM-as-judge scoring across accuracy and reasoning quality. Shipped Entrata's first production AI pipeline (PyTorch + OpenAI API) across 1.6M real financial records with output validation, quality gates, and compliance controls. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

Entrata

Jan 2017 – Aug 2024

Trainee → Associate Engineer → Software Engineer

- Pioneered Entrata's **first production AI pipeline** (Python, PyTorch, OpenAI API) across 1.6M real financial records — full lifecycle from research through output validation, quality gates, and production deployment; same evaluation-first discipline Fieldguide applies to audit AI.
- Built and owned **20+ production modules** (Python, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints for 100K+ monthly active users and 1M+ property records — 99.5% uptime over 6+ months; cut deployment times 40%, query execution 50% via Docker containerization.
- Achieved **100% PHI compliance** on AWS S3 migration (Libsodium-encrypted pipelines, zero security incidents); improved SQL performance 30% across 200+ queries.
- Automated compliance workflows (LIHTC, Section 8/HCV, HUD) with zero violations — **eliminated 80% of manual workload** via Python/PL-SQL automation across 100K+ unit portfolios.

George Mason University

Aug 2024 – Aug 2025

Research Associate — Statistical Modeling · Anomaly Detection

- Built regression models ($R^2 = 0.977$) in Python and R across 190+ countries; applied Isolation Forest to automate anomaly detection across 3 research pipelines — surfaced anomalies manual review missed.

PROJECTS

AgentProbe: Unified Evaluation & Observability Platform for Agentic AI

Open Source

Python · LangGraph · LangChain · FastAPI · ChromaDB · LLM-as-Judge · Anthropic · OpenAI · Groq · Ollama · React · Docker · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- Built a **unified 5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval quality, completeness, compliance) serving as the single source of truth for agent quality — CI/CD gates block failing runs before production; automated pipelines evaluate agent behavior on every commit.
- Delivered **trace-level agent observability** via React dashboard — per-run metrics, retrieval quality, regression drift, latency, and failure mode surfacing across concurrent agent test suites; same observability stack Fieldguide is building for audit workflows.
- Built a 3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) with tool calling, inter-agent state handoffs, and **multi-provider LLM routing** (Anthropic, OpenAI, Groq, Ollama) — 15-endpoint FastAPI backend; comparison framework measuring effectiveness, consistency, and latency across model versions.

LeaseI: Agentic RAG Pipeline with Pass@k Evaluation Harness

Python · LangGraph · LLaMA · Transformers · SQLite · reportlab — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built a production-grade agentic RAG pipeline over 1,240 residential lease PDFs — **93.6% precision, 87.9% F1** across 10 compliance rules; **pass@k eval harness** (pass@1 81.7%, pass@3 88.3%) with multi-trial sampling, regression gate, and LLM-as-judge scoring of both accuracy and reasoning quality; deterministic rules engine keeps audit decisions fully traceable.

AI on FHIR: NLP Pipeline for Regulated Clinical Records

Flask · spaCy · BERT · Transformers · Embeddings · LLMs · React · Docker — github.com/venkatalakshmiparimala/AI_on_FHIR

- Built a HIPAA-compliant NLP platform translating natural language queries to FHIR APIs using spaCy, BERT, and semantic embeddings — **RBAC**, ~60% reduction in retrieval time; demonstrates production AI in a compliance-constrained domain analogous to Fieldguide's audit environment.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia

B.Tech Computer Science & Engineering — Sree Venkateshwara College of Engineering — 2013 – 2017

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, PL/SQL, R, Bash, HTML/CSS

Eval & Observability: LLM-as-Judge, LLM Evaluation, Eval Frameworks, Pass@k Harnesses, Hallucination Detection, Retrieval Quality Metrics, Regression Gates, CI/CD Quality Gates, Agent Tracing, Agent Observability, LangSmith (familiar), Langfuse (familiar)

Agentic AI & RAG: LangGraph, LangChain, Multi-Agent Orchestration, Tool Calling, RAG Pipelines, ChromaDB, FAISS, Vector Databases, Semantic Search, Embeddings, FastAPI, REST APIs, OpenAI API, Anthropic API, Groq, Ollama

ML & NLP: PyTorch, Scikit-learn, spaCy, BERT, Transformers, XGBoost, Pandas, NumPy, Hugging Face

Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, GitHub Actions, PostgreSQL, MySQL, SQLite, React, Git, Linux, Agile/Scrum (120+ sprints)